



Tshwane University of Technology

We empower people

SURNAME & INITIALS: MANALA LL

STUDENT NUMBER: 230325898

COURSE: COMPUTER SYSTEMS ENGINEERING

SUBJECT: PD135D House Wiring

DATE: 17 April 2026

Table of contents

Contents

1. Introduction
 - Aim of the project.
 - Scope of the project.

2. Design
 - List of appliances and their consumption.
 - Selection and sizing of components.
 - House plan layout.

3. Testing
 - List of tools used when wiring
 - Safety and risk management
 - Testing/Demonstration

4. Conclusion

5. Appendices.
 - Bill of materials
 - Other information

1. INTRODUCTION

- AIM
 - The aim of this project is to design a safe and efficient electrical wiring diagram for a 3-room house in compliance with SABS (SANS 10142-1).

- PROJECT SPECIFICATIONS.

The wiring design includes:

- A distribution design from the main supply to the final circuit.
- Lighting circuits for all rooms using LED bulbs (2 for the living room, 1 for the kitchen, and 1 for the bedroom).
- Socket outlet circuit for in-house appliances.
- Dedicated circuit for the electric geyser.
- Dedicated circuit for a stove.
- Earth leakage protection for all socket circuits.

2. DESIGN.

- List of all appliances and power ratings
 - Electrical components
 - Electrical socket – 16A
 - Distribution box.
 - Main Breaker – 63A
 - Residual current breaker – 63A
 - Geyser breaker - 63
 - Stove breaker – 32A
 - Light breaker – 10A
 - Plug breaker – 20A
 - Home appliances.
 - TV – 100W
 - Stove – 3000W
 - Fridge – 150W
 - Geyser – 3000W
 - LED Lights(4x) -18W
 - Meter Box.
- Selection and sizing of components.

$$P = V \cdot I.$$

Where:

I = Current in Amperes (A)

P = Power in Watts (W)

V = Voltage in Volts (V) = 230 V (South African nominal)

- Lighting circuit calculation.

SANS 10142-1 Section 6.2.6.6 requires that the current-carrying capacity of conductors for LED and CFL lighting circuits be based on double the sum of the lamp load(s) in watts. This accounts for harmonic currents generated by these fittings.

Total power = $18 \times 4(\text{rooms}) = 72\text{W}$

$72\text{W} \times 2 = 144\text{W}$.

$I = 144\text{W}/230\text{V} = 0.62\text{A}$

Therefore, a 10A breaker will work.

Cable selection:

1.5 mm² PVC cable rated at 10 A in conduit.

- Socket Circuit Calculation

SANS Socket standard is rated for 16A

20A Breaker.

Cable selection:

2.5 mm² PVC cable rated at 16-20 A in conduit

- Geyser Circuit Calculation

$P=3000\text{W}$

$I = 3000\text{W} / 230\text{V} = 13.04\text{A}$.

Breaker sizing per SANS 10142-1:

$I_n \geq I \times 1.25$ (safety margin for continuous loads)

$I_n \geq 13.04\text{ A} \times 1.25 = 16.3\text{ A}$

Selected: 20 A MCB (next standard size)

Cable selection:

4.0 mm² PVC cable rated at 25 A in conduit

- Stove Circuit Calculation

Power = 3000 W

$I = 3000\text{ W} / 230\text{ V} = 13.04\text{ A}$

Breaker sizing:

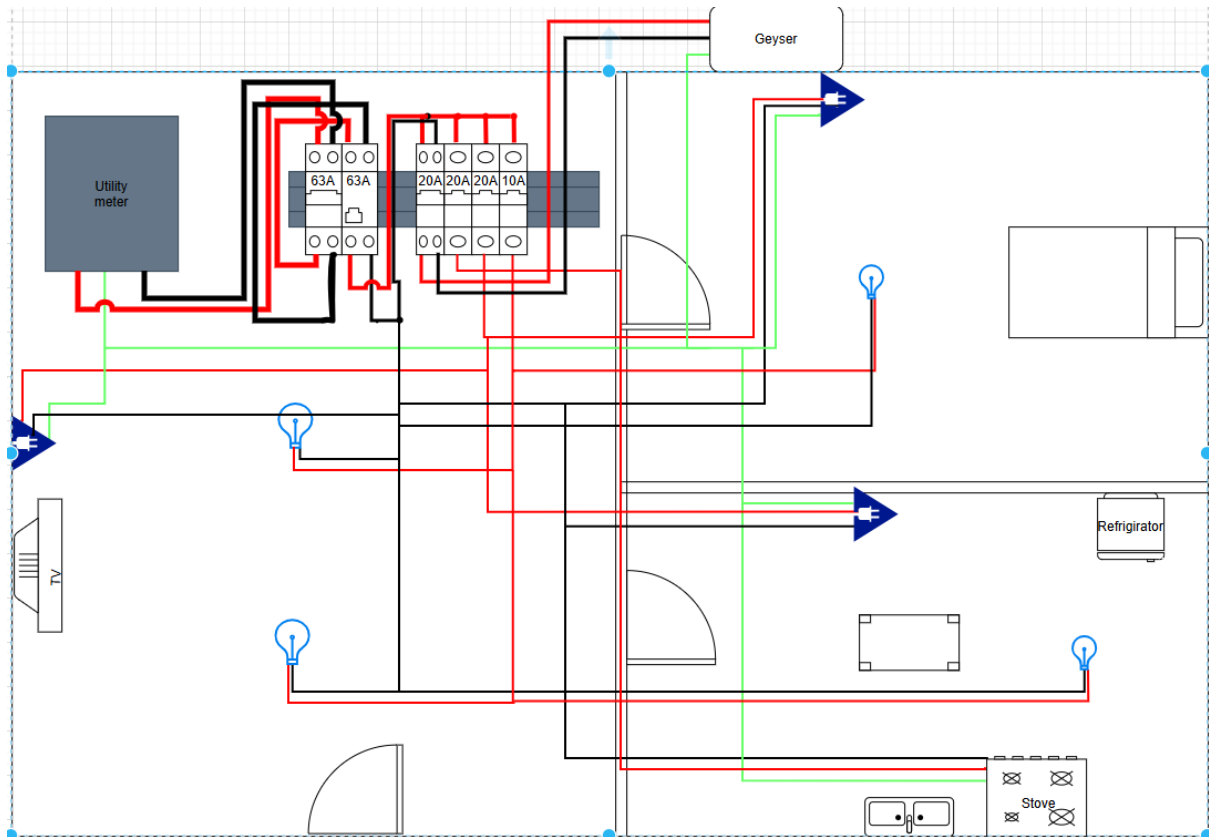
$I_n \geq I \times 1.25 = 13.04\text{ A} \times 1.25 = 16.3\text{ A}$

Selected: 20 A MCB (Next standard size)

Cable selection:

4.0 mm² PVC cable rated at 25 A in conduit

- House plan layout.



So, in this layout, we have a 3-roomed house (bedroom, living room, and kitchen) powered from a distribution box. And that is powered by the Earthed utility meter. The bedroom has a single light and a single wall socket, both wired in parallel with their respective circuits. The living room has 2 lights and 1 plug connected to a 10A and a 20A breaker, respectively. The configuration of the kitchen is the same as the bedroom, with the light and plug wired in parallel with their circuits, then to a 10A and a 20A circuit breaker. The stove has its own dedicated circuit wired to a 20A breaker, and the geyser also has its own circuit connected to a 20A circuit breaker. The conductor selection is as follows: Green – Earth, Red – Live, Black – Neutral.

3. TESTING

List of tools used when wiring.

- Digital Multimeter - Voltage, current, and resistance measurement.
- Flat head screwdriver - Used to tighten or loosen screws with flat heads.
- Star head screwdriver - Like the flat screwdriver but designed for screws with star-shaped heads.
- Spirit Level - Ensure switches and sockets are level
- Side cutter - Cable insulation removal without damage
- Gloves – For safety and to maintain grip on tools.
- Impact drill – Used to drill holes in boards or walls.
- Hacksaw – Used to cut conduits.
- Fish tape – used to insulate spliced wires and exposed wires.

Safety and Risk Management.

- Personal Protective Equipment.
 - Safety glasses – protect eyes from sparks and debris.
 - Insulated gloves - Protects hands from electric shock and accidental contact with live wires.
 - Safety boots – Protect the feet and offer electrical insulation.
 - Ear protection – Used when working near loud equipment to protect one's ears.
 - Hard Hat - Head protection from falling objects.
 - High visibility vest - Ensures visibility in workplaces with multiple workers or moving equipment.
 - Dust Mask – Protects against dust or toxic fumes when drilling holes or installing conduits.

- Common Faults.
 - Overcurrent
 - Breaker trips rapidly with a burning smell and can be caused by an overloaded circuit or a short circuit.
 - Prevention: Identify and remove fault, redistribute load, repair/replace cable.
 - Open circuit
 - Break in the circuit preventing current from flowing.
 - Prevention: Repair broken conductors and tighten connections or remake joints.
 - Incorrect breaker ratings.
 - Using an underrated breaker or overrated breakers can lead to ineffective protection against current overload.
 - Solution: Do load calculations to select the right breaker.
 - Poor cable management
 - Exposed or tangled wires can cause short circuits or arcing.
 - Solution: Use cable ties and conduit pipes to keep the wires organised.
 - Polarity reversal.
 - Live and neutral wires are incorrectly installed.
 - Solution: install the live and neutral wires in the proper orientation at the affected point.
 - Earth fault.
 - Current flows from live conductor to earth due to insulation failure.
 - Solution: Locate fault using an insulation tester, replace the damaged cable, and ensure proper earthing

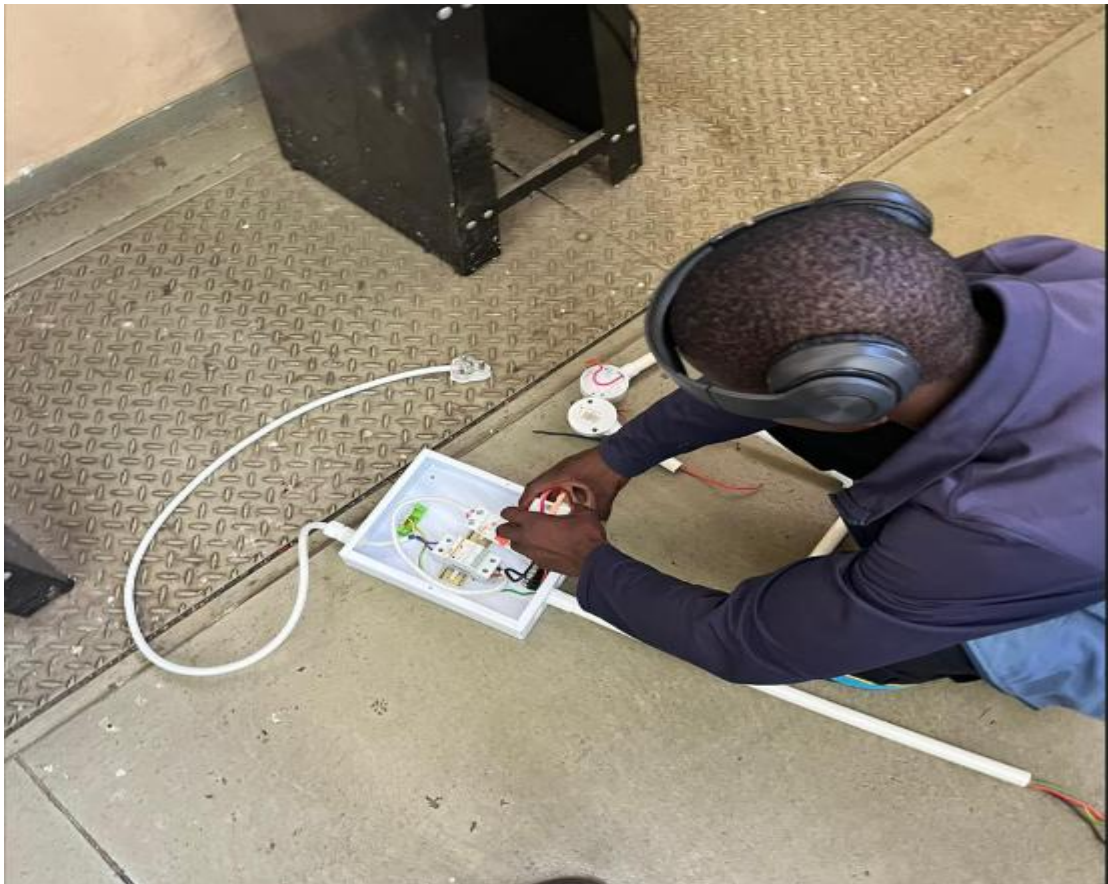
Testing/Demonstrating

Before testing, we had to disconnect the given LAB circuit, then reconnect it.

1. Disconnect the circuit.



2. Circuit disconnected.



3. Reconnecting the circuit with the team.



4. Circuit Reconnected



5. Circuit test.



4. CONCLUSION.

The house wiring design project was completed in accordance with the requirements of SANS 10142-1. The design ensures safe, efficient, and practical electrical distribution for a three-room residential dwelling. All appliances were carefully selected, and their power ratings were used to determine the total load demand of the installation. Based on these calculations, appropriate circuit breakers, cable sizes, and dedicated circuits were selected to ensure compliance with safety standards and operational reliability.

The distribution board was designed to accommodate both general and high-load circuits, with specific consideration given to appliances such as the stove and geyser, which require dedicated protection. Proper load balancing and circuit grouping were implemented to prevent overloading and to improve overall system performance. The wiring layout and house plan were developed to ensure logical routing of cables and ease of installation, while also maintaining compliance with regulatory requirements.

Several challenges were encountered during the design process:

- **Diversity factors:** Selecting a realistic value required balancing safety and practicality. A factor of 0.7 was used.
- **Breaker sizing:** Initial assumptions were corrected through proper calculations to ensure correct protection.

- **Conduit sizing:** Required careful use of standard tables to ensure compliance with fill limits.

Overall, the project demonstrated the importance of accurate calculations, adherence to standards, and proper design practices in electrical installations. The final design meets all functional and safety requirements and reflects a practical understanding of residential wiring systems.

Recommendations.

Surge Protection

Install a surge protection device (SPD) at the main distribution board to protect sensitive electronic equipment from voltage transients caused by lightning strikes or switching operations. This is particularly important for the television, decoder, and charging devices in the living room and bedrooms.

Energy Efficiency

Consider upgrading all lighting to high-efficiency LED fittings with a minimum efficacy of 100 lumens per watt. This will reduce the lighting load and associated energy costs. Additionally, the installation of motion sensors in the passage and bathroom could further reduce unnecessary energy consumption.

Future Expansion

The 8-way distribution board includes one spare way (Way 8) for future circuit additions. It is recommended that this spare capacity be reserved for potential air conditioning installation or an additional dedicated circuit for a home office. When expanding, ensure that the main breaker rating and supply capacity are not exceeded.

Regular Maintenance

Schedule annual electrical inspections by a registered electrician to verify the continued integrity of the installation. Attention should be paid to:

- Tightness of terminal connections (torque check)
- Condition of earth bonding
- Functionality of earth leakage protection
- Signs of overheating or insulation degradation
- Operation of isolators and switches

Documentation

Maintain a complete set of as-built drawings and test records on-site for future reference. This documentation should include:

- Updated floor plans showing actual cable routes
- DB schedule with circuit descriptions
- Test certificates and COC
- Appliance manuals and warranties
- Maintenance log

Safety Awareness

Ensure all occupants are aware of the location of the main switch, earth leakage test button, and emergency procedures. Post emergency contact numbers (electrician, fire department, ambulance) near the distribution board.

APPENDICES.

Bill of materials

Item No.	Description	Specification	Quantity	Estimated Cost (ZAR)
1	Distribution Board (DB)	8-way surface mounted	1	R800
2	Main Circuit Breaker	63 A MCB	1	R250
3	Earth Leakage Unit	63 A, 30 mA RCCB	1	R600
4	Lighting Circuit Breaker	10 A MCB	1	R120
5	Socket Circuit Breaker	20 A MCB	2	R240
6	Geyser Circuit Breaker	20 A MCB	1	R120
7	Stove Circuit Breaker	20 A MCB	1	R120
8	Electrical Sockets	16 A double plug	3	R180
9	Light Switches	Single switch	4	R120
10	LED Lights	18 W bulbs	4	R200
11	PVC Cable (Lighting)	1.5 mm ² twin + earth	30 m	R300
12	PVC Cable (Sockets)	2.5 mm ² twin + earth	40 m	R600
13	PVC Cable (Geyser/Stove)	4 mm ² twin + earth	30 m	R700
14	Conduit Pipes	PVC conduit (20 mm & 25 mm)	20 lengths	R400
15	Conduit Accessories	Bends, couplers, saddles	Set	R200
16	Meter Box	Standard utility meter box	1	R500
17	Earth Rod & Clamp	Copper earth rod	1	R300
18	Insulation Tape	Electrical tape	2	R40
19	Cable Ties	Plastic ties	Pack	R60

